Section 1 : General and Combinational Circuit Design (10 marks)

1. Given 3 bits unsigned integer **A,B** designing a circuit that provide the output of absolute value of the difference **|A-B|**

2. Show designing process and draw the circuit of **8 to 3 Priority Encoder** with active low outputs and active high enable signal

3. Explain difference between 3 types of shifting operation

Section 2: Data representation & sequential circuit design

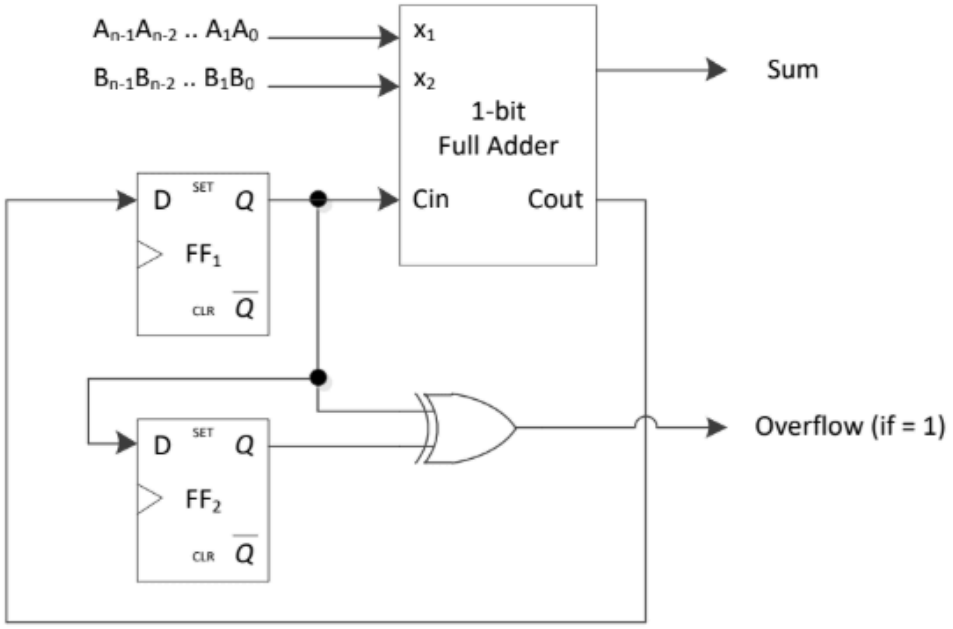
1. Given the floating point format (1 sign bit, 10 exponent bits, 21 fraction bits)
   What is the **biggest** positive number and **smallest** negative number that can be represented in this format(in decimal).

2. Design 3 bits multi-function register with this following modes

s1	s0	mode
0	0	Decrement by 2
0	1	(x2 - 3) mod 8
1	0	Parallel input
1	1	Circular shift right

Section 3: Arithmetic Circuit & Arithmetic operation

1. Explain how to modify the serial adder circuit to do subtraction operations A - B when A and B is 5 bits, A = -3 , B = 6



Section4: RISC control unit design & runtime analysis

1. Design and Draw a **LOAD** control unit of register
    a. Register AR
    b. Register AC

2. From this main memory, assume first line start at address 0x00

    a. Write the HEX code corresponding to content of each address

Symbolic code	HEX code
LDA 00B	
STA 00C	
LDA 00D	
STA 00E	
CLA	
ADD 00C I	
ISZ 00C	
ISZ 00E	
BUN 005	
STA 00F	
HLT	
0000 0000 0000 0000	
0000 0000 0000 0000	
1111 1111 1111 0110	
0000 0000 0000 0000	
0000 0000 0001 0010	
0000 0000 0000 0011	
0000 0000 0000 0100	
0000 0000 0000 0101	
0000 0000 0000 1100	
0000 0000 0000 0111	
0000 0000 0000 1000	
0000 0000 0000 1001	
0000 0000 0000 1010	

b. Find the value of AC and flip-flop E after 40 clocks (using the code provided in the previous page)

10:48 Fri 31 Oct
96%
3
CA midterm | Midterm_Moc... | CA Midterm | CA midterm_p... | 3

Instructor: Thanadol PRITRANAN Total Marks: 50

Date: 11 ตุลาคม 2564 Hours: 1300 - 1440

Instruction: *Answer all questions and submit to our Google Classroom*

Outline of Midterm Examination:

Questions 1 & 2/ 25 minutes/ 12 & 10 marks (1325-1350) Control Circuit Design

- 5 min break –

Questions 3/ 20 minutes/ 16 marks (1355-1415) Runtime Analysis

- 5 min break –

Questions 4/ 20 minutes/ 12 marks (1420-1440) Interrupt Mechanism

(16 คะแนน) Runtime Analysis

(16 คะแนน) **Q3.** กำหนดให้ x คือค่า 3 digits หลังของรหัสนักศึกษา mod ด้วย 5 (เช่น นักศึกษาที่มีรหัสนักศึกษาลงท้ายด้วย 456 ค่า x จะเท่ากับ 456 mod 5 .. x = 1) จากข้อมูลที่กำหนดให้ สมมติขั้นตอนการ fetch คำสั่ง CLE จาก address 100H เริ่มที่ t_m, ค่าใน register AC เมื่อสิ้น clock t_{m+35+x} จะเท่ากับเท่าไหร่ และให้อธิบายสั้นๆ ถึงแนวทางการหาคำตอบ

Main Memory

100_H	CLE
	LDA 113_H
	CIR
	STA 113_H
	SZE
	BUN 107_H
	BUN $10B_H$
	LDA 112_H
	ADD 114_H
	STA 114_H
	CLE
	LDA 112_H
	CIL
	STA 112_H
	ISZ 111_H
	BUN 111_H
	HLT
	$FFF8_H$
	$000F_H$
	$000B_H$
	0000_H

Main Memory

100_H	CLE
	LDA 113_H
	CIR

1 of 3

Given

Main memory

CLE	100_H
LDA 104	101_H
ADD 103 I	102_H
0105_H	103_H
$AABB_H$	104_H
$CCDD_H$	105_H

SC = 0

PC = 102 H

AR = 104_H

IR = 2104_H

DR = 5555_H

AC = $AABB_H$

E = 0

After execute one instruction.

What is this instruction? :

How many clock cycles instruction uses? :

value of SC :

PC :

AR :

IR :

DR :

AC :

E :